Determine

Dumb glasses, a capable phone, and a calibrated warning about what is directly ahead of you.

Architecture and prototype contract

Status: proposed design. Nothing here is built, clinically validated, or cleared as a medical device. It does not replace a white cane, a guide dog or orientation-and-mobility training.

WEARABLE
Two dumb boards: an ESP32-CAM that streams JPEG, and a XIAO ESP32-C3 carrying the time-of-flight array, stereo bone-conduction cues and two haptic motors. No AI on the frame.

PHONE
Does all perception, decides which hazard to report, runs calibration and the user interface. Android first.

USERS
Blind and low-vision travellers, including people with unilateral or asymmetric hearing loss.

TARGET
HackTX 26, Oct 24–25 2026, UT Austin. Build on site within the event rules.

> **Pitch.** When you reverse a car, your view is limited, so the car gives you parking sensors. People who are actually blind or have low vision mostly do without a head-level equivalent. Determine is that sensor for a walker: it beeps faster as an obstacle gets closer, places the warning on the left, centre or right, and calibrates its loudness so both sides are heard equally, even with hearing loss in one ear.

How to read this contract

This document fixes *interfaces and invariants*. It does not claim that the hardware works or that the design has been field-tested. **MUST**, **MUST NOT**, **SHOULD** and **MAY** state obligations on the proposed implementation only. Every claim carries a tag:

Tag	Meaning
FACT	Read from a primary or vendor source, cited with [n]. Vendor statements describe the vendor's own product, not an independent check of Determine.
DESIGN	An engineering decision this document makes, with its rationale stated.
INFERENCE	Arithmetic or reasoning from facts, not measured. A gate must confirm it before anyone repeats it as a result.
PROPOSED	A starting threshold to tune in a pilot. It is not a standard, a regulation or a measurement.
GATE	Measured evidence required before a capability may be claimed or the next layer built.

Most vendor prices below were checked 24 Oct 2026 and exclude tax and shipping. SparkFun's ToF price/stock and Adafruit's bone transducer price/out-of-stock status were rechecked 18 Oct. Both Micro Center product pages (XIAO ESP32-C3 3-pack; ESP32-CAM pair) blocked automated reads (HTTP 403), so their prices and local availability are unverified. The sourcing chapter separates verified facts from calls you need to make. Layout follows the reference *MisaOS-Architecture.pdf*. Determine is an independent project.

CONTENTS

Contents

1 Mission, guarantees and non-goals
2 Evidence base: travellers, hearing and cues
3 Landscape and positioning
4 System architecture and seams
5 Wearable hardware
6 Links: Wi-Fi, USB, BLE, LE Audio, Auracast
7 Perception and hazard policy
8 The cue language
9 Calibration for asymmetric hearing
10 Safety, privacy and regulatory boundary
11 HackTX 26 build plan
12 Sourcing in Houston and bill of materials
13 Verification gates and failure catalogue
14 Roadmap after HackTX
15 Architecture decision records
16 Source register

Decisions at a glance

Question	Decision	Why
Where does computing happen?	On the phone. Two dumb boards on the glasses capture, transmit and play back. DESIGN	Cheap, replaceable frame. The phone already has an NPU, a battery and accessibility settings. One place to update models.
Two cameras?	No, for v0. Use one RGB camera plus an 8×8 time-of-flight (ToF) array. DESIGN	Stereo depth needs synchronised capture and calibration. ToF measures distance whatever the obstacle class, which is what head-level hazards need.
Camera on the cue board?	No. A separate ESP32-CAM streams JPEG over the same hotspot. DESIGN	The XIAO ESP32-C3 has no camera, and the esp32-camera driver supports ESP32, S2 and S3 only [13][59]. A second Wi-Fi producer is the price of keeping the cue board simple.
Camera link	JPEG from the ESP32-CAM over the phone's Wi-Fi hotspot. USB UVC only after a per-phone check. DESIGN	BLE cannot carry useful video [15]. Android external-camera support depends on the phone maker [10].
Urgent cue path	The phone sends a cue ID and the cue board plays a stored earcon and haptic pattern. No A2DP. DESIGN	A2DP typically adds 100–200 ms [16]. Reflex cues need to arrive in tens of milliseconds.
Conveying direction	Redundantly: calibrated audio side, haptic side, and an optional spoken word. Never stereo panning alone. DESIGN	Bone-conducted sound crosses the skull [26]. People with single-sided deafness cannot use level differences [28][29].
Calibration	Per session, per side: detection threshold, then a loudness balance, with a hard ceiling. DESIGN	This is the user's core requirement. Coupling and hearing differ by person and by how the frame sits.
Cloud	Never in the hazard loop. Scene description is optional and labelled "may be wrong". DESIGN	Network loss must not silence warnings. A 2026 study found multimodal-LLM answers wrong 22.2% of the time [30].
Meta and AWS	Meta DAT is an alternative frame source behind the same interface. AWS is user-requested description only: a Graviton2 Lambda API plus SmolVLM on Inferentia2 is the design target, untested (§7.3). DESIGN	Shows real integration without weakening the cost thesis or the offline invariant.

CHAPTER 1

Mission, guarantees and non-goals

1.1 The precise product

Determine is a **secondary electronic travel aid** (ETA). It adds a near-field warning layer, focused on chest- and head-level obstacles, on top of the traveller's primary aid. The 2026 *Scientific Reports* study defines primary aids as the white cane and guide dog, and ETAs as supplementary tools [2]. A long cane covers the ground and "does not protect users from obstacles at the upper body and head levels" [2]. In a survey of 307 blind travellers, 13% had a head-level accident at least monthly, 86% of head-level accidents happened outdoors (mostly tree branches), and 23% of head-level accidents had medical consequences [3]. FACT

The wearable is deliberately "dumb": it has sensors, radios and actuators, but it makes no decisions. The phone app is the processing unit. It decides what counts as a hazard, which single cue to play and how loud each side should be. DESIGN

1.2 Guarantees

1. **G1 Fail loud.** If the camera, ToF, link, phone app or wearable power fails, the system targets an audible or tactile "system unavailable" notification through a surviving output within ≤ 1 s PROPOSED. If both boards lose power, the phone uses its own speaker and vibration; the dead frame cannot play an earcon. Silence means "nothing to report *from a working system*", never "path clear".
2. **G2 Unknown is not clear.** An empty detection list never produces a reassurance cue. Determine has no "all clear" sound.
3. **G3 Offline hazard loop.** Capture → perception → decision → cue runs with no internet connection.
4. **G4 Calibrated, capped output.** Per-side gain comes from the wearer's own calibration and is clamped by a ceiling that no setting can exceed.
5. **G5 Redundant direction.** Every directional cue is carried by at least two channels the wearer has confirmed during calibration that they can perceive.
6. **G6 Bounded vocabulary.** Hazard cues come from a small, learnable set (Chapter 8). Free-form speech never pre-empts a hazard cue.
7. **G7 Privacy by default.** Frames stay in phone memory. Images are not retained or uploaded unless the user explicitly turns on a separate feature.

1.3 Explicit non-goals

Not promised	Reason
Replacing the cane, guide dog or O&M training	Primary aids provide ground contact and legal recognition. ETAs supplement them [1][2].
Drop-offs, curbs, stairs down, holes	Forward-facing head-level sensing does not see the ground ahead reliably. No claim without per-class test data.
Street crossing or traffic judgement	Safety-critical. Out of scope.
Glass doors, thin wires, wet or very dark surfaces	Known failure modes for camera and ToF sensing, listed as "cannot see" in the app.
Restoring directional hearing	Bone conduction and rerouting do not restore localisation in single-sided deafness [28].

Not promised	Reason
A certified sound-exposure dose from bone conduction	No verified standard maps bone-conduction transducer output to air-conduction dose limits. The prototype caps output but cannot certify it.
Unsupervised public trials during HackTX	Demos run on a staged indoor course with a sighted spotter.

1.4 Claim classes

Class	Meaning	Example
C0	Measured on the prototype, with the log kept	"p95 capture-to-cue latency was 212 ms over 120 trials." Only after the test exists.
C1	Vendor or primary-source fact about a part	"The VL53L5CX module has 8×8 zones and a 4 m range" [20].
C2	Design target	"Unavailable cue within 1 s."
C3	Research direction	"Deliver cues to LE Audio hearing aids."

The pitch, slides and Devpost entry **MUST** label class C2 and C3 statements as such. Presenting a C2 target as a C0 result is a release blocker.

CHAPTER 2

Evidence base: travellers, hearing and cues

2.1 What a real ETA trial shows

Pittet et al. (2026) tested 13 blind and low-vision participants on an indoor corridor with floor, waist and head-level obstacles. They compared cane alone with cane plus NOA (a camera-based harness) and cane plus BuzzClip (ultrasonic) [2]. FACT

Measure (mean)	Cane only	Cane + NOA	Cane + BuzzClip
Body contacts with obstacles	2.92	1.62	3.54
Cane contacts	4.69	2.62	3.15

What this means for Determine:

- **Precision beats volume.** Participants rated NOA higher on precision and on upper-body detection. The BuzzClip's vibrations were criticised as too weak and too frequent, and gave no improvement in incidents [2].
- **Learnability matters.** 4 of 13 participants had trouble understanding NOA's spatialised warning sounds [2]. Determine therefore uses a small set of cues, trained during onboarding, with direction carried redundantly.
- **Short, precise beats generative.** For finding objects, every participant preferred concise output such as "Person to your left at 11 o'clock, 3 metres" from a local deep-learning model over generative-AI descriptions [2].
- **Test with the real population.** Of 140 ETA studies reviewed, only 29% included blind or low-vision participants [2]. A blindfolded sighted tester does not validate Determine.

2.2 The glasses premise, without stereotypes

Some blind people wear dark glasses, but many do not. Blindness is a spectrum, and many legally blind people use their remaining vision. Determine therefore offers clear, tinted and clip-on configurations, and its sensors mount on the frame front so that lens tint does not affect sensing. DESIGN A pair of glasses is a natural place for the hardware: it sits at eye level, it points where the head points, and it already rests on the temples, where bone conduction and haptics make contact.

2.3 Hearing: two different problems

Asymmetric hearing loss

Both ears hear, but one hears less. Per-side gain can equalise *perceived* loudness, which is exactly what the user asked for. Calibration is the right tool here. INFERENCE

Single-sided deafness (SSD)

One cochlea does no useful work. Every left or right channel ends up at the same working cochlea. In 28 unilaterally deaf subjects, sound crossing the skull lost only about 0–10 dB, depending on frequency, with roughly 40 dB variation between people [26]. Left/right audio therefore cannot carry direction, so haptics and spoken words must. FACT → DESIGN

- Bone-conduction headsets lateralise sound much like headphones in a small study. Even a 20 dB level difference gave only about 75% correct lateralisation [27]. Stereo bone conduction is useful, but it is not reliable enough to be the only carrier of direction. FACT

- In one study, a contralateral-routing bone-conduction device did not improve localisation for people with single-sided deafness [28]. Monaural listeners rely heavily on head-shadow level cues [29]. FACT
- Audio played through bone-conduction headphones increased localisation errors for real environmental sounds [52]. Determine keeps cues short and sparse so it does not mask traffic and voices. FACT → DESIGN

2.4 Auditory-display findings used in Chapter 8

- **Earcon guidelines** (Brewster et al.): vary timbre, register and rhythm, and use musical timbres rather than sine waves. Make rhythms differ substantially, including in note count. Keep pitch roughly within 125–150 Hz to 5 kHz. Notes should last at least 0.0825 s, at most about 6 notes per second, with about 0.1 s between serial earcons. Keep intensity 10–20 dB above threshold. Do not rely on register or intensity alone for absolute identification [24]. FACT
- **Spearcons**, which are sped-up spoken words, beat earcons and auditory icons on menu-navigation efficiency and accuracy, and are learned about as fast as speech [25]. FACT
- **Safe listening:** WHO lists 80 dB as safe for up to 40 hours a week and 90 dB for only 4 hours. It recommends keeping devices at or below 60% of maximum volume [4]. The WHO-ITU standard defines exposure modes for adults and sensitive users [5]. FACT

CHAPTER 3

Landscape and positioning

Product	Form	Relevant capability	Price (source, 24 Oct 2026)	Limitation relative to Determine's goal
Ray-Ban Meta (Gen 3)	Camera and open-ear speaker glasses	Meta AI scene description, Be My Eyes volunteer calls. Developer access via Wearables Device Access Toolkit (DAT) 1.0 [6]	From $449 [42]	AFB calls it an accessibility tool "only by coincidence" and reports hallucinated object positions [9]. Air-conduction speakers, not bone conduction. No haptics. Gen 3 is not named in the DAT supported-device list [6]
Envision Ally Solos	Camera glasses, phone-tethered	Reading, scene description, faces	$699 shop listing [43]	Description-focused, not near-field hazard cues
OrCam MyEye 3 Pro	22.5 g clip-on camera	Reading, faces, products	$4,250 reseller [44]	Reading-centric. Not a navigation device
WeWALK Smart Cane 2	Smart cane handle	Chest and head-level obstacle detection with vibration, navigation app	Not on the static official page [45]	Occupies the cane hand. Closest neighbour in function
biped NOA	Shoulder harness, three depth cameras	Spatialised obstacle audio sent to bone-conduction headphones [2]	$4,990 reseller [46]	8 of 13 trial users found it heavy or bulky [2]. Expensive
.lumen glasses	Head-worn, forehead haptics	Haptic path guidance	€9,999 [47]	Ships only in Romania per its order page. Claims such as "better than the guide dog" are unvalidated marketing
NaviLens	Markers plus a free app	Long-range marker reading in transit systems	Free [48]	Needs installed infrastructure. Wayfinding, not obstacle avoidance
Seeing AI / Lookout	Phone apps	Text, scene and object description	Free [49][50]	Handheld. Lookout's Explore mode is beta and "less accurate" [50]
Microsoft Soundscape	Retired spatial-audio app	3D audio beacons	Open-sourced [51]	Microsoft says it is "not … for use as a mobility aid" [51]

3.1 Where Determine sits

None of the products found combines all four of the following: a low-cost glasses form, phone-side perception, bone-conduction *and* haptic output, and explicit calibration for asymmetric hearing. No ETA product or paper found addresses unilateral hearing loss in its cue design. That is an inference from absence of evidence, not a verified gap. INFERENCE

Not "AI glasses for the blind"

That framing is crowded and invites overclaiming. Determine is a *parking sensor for walking*, whose output follows a defined set of rules.

Commodity hardware, unpriced build

The phone does perception; the frame uses two commodity boards. Only $88.35 of line items have quoted prices, including out-of-stock transducers. Board packs, drive extras and power remain unpriced (Chapter 12). Do not quote a frame total yet.

Hearing-aware by construction

Calibration, a mono-safe mode and haptic direction are core features, not settings bolted on later.

Claims to avoid in the pitch: "replaces the cane", "detects stairs, curbs or traffic", "makes navigation safe", "restores hearing", "first obstacle sensor for the blind" (WeWALK and NOA already exist), "uses Auracast" (unless it was actually built and demonstrated).

CHAPTER 4

System architecture and seams

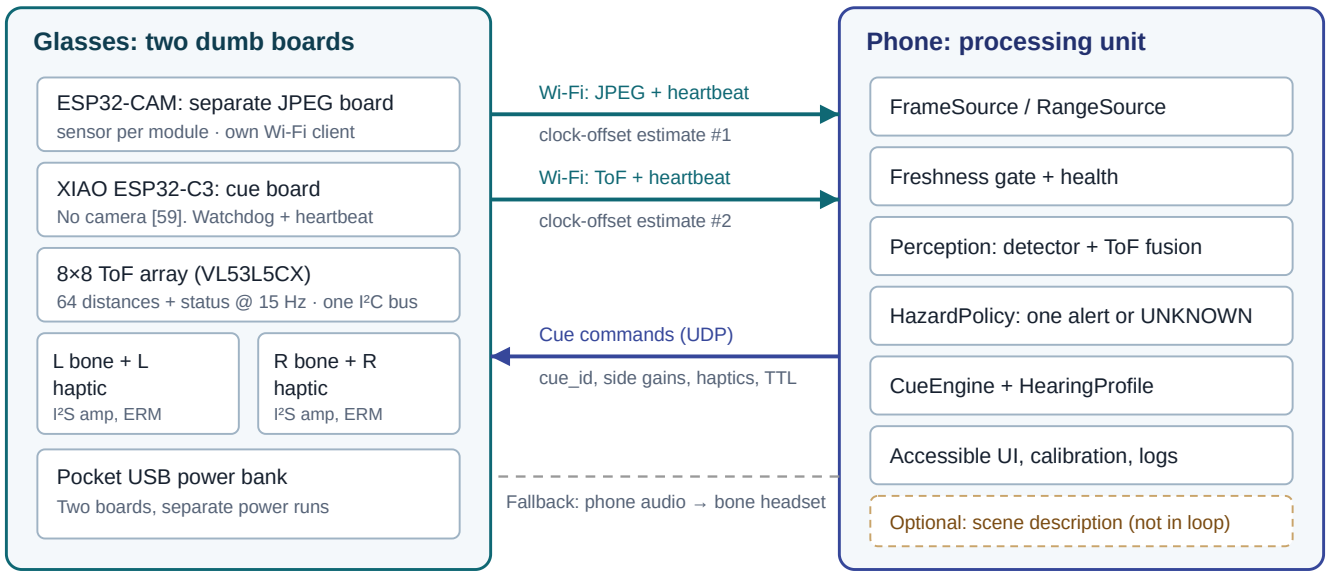


Figure 4.1. Topology: two independent Wi-Fi producers and no board-to-board bus. The phone is the only integrator — it aligns each stream by that board's estimated clock offset and receive age, and discards stale input. The wearable never decides what is hazardous. Cue commands carry a time-to-live (TTL), so a stale command is dropped rather than played late.

4.1 Module seams

Each seam exists only where at least two implementations are real or planned, following the "two adapters means a real seam" rule. DESIGN

Module	Interface	Adapters	Invariants and error modes
FrameSource	`next(): Frame{seq, tCapture, w, h, jpeg}` or `Unavailable(reason)`	ESP32-CAM Wi-Fi · USB UVC · Meta DAT · phone camera (Tier 0)	Sequence numbers increase monotonically. Clock offset is estimated per board by PING/PONG; receive age is tracked. Stale input is discarded. Disconnection is explicit, never an empty frame.
RangeSource	`next(): RangeGrid{seq, t, mm[64], status[64]}`	VL53L5CX on the XIAO ESP32-C3 · none (camera only)	An invalid zone means UNKNOWN, never "far". Same per-board offset and staleness rules as frames.
Perception	`evaluate(frame, range?): Observations`	MediaPipe detector · plus optional depth model	Every observation carries its source age and confidence. No metric distance from bounding-box size alone.
HazardPolicy	`select(obs, motion): Alert | Unknown | Quiet`	Single rule-based implementation (internal seam for tests)	At most one alert at a time. Entry and exit use hysteresis. Never emits "clear".

Module	Interface	Adapters	Invariants and error modes
CueSink	`present(alert, profile): Ack | Fault`	Glasses playback over UDP · phone audio to headset · phone vibration (non-directional)	Gains are clamped by the profile ceiling. A missing acknowledgement escalates to UNAVAILABLE.
HearingProfile	`calibrate(sink): Profile` · `gains(cue): (gL, gR)`	One per output device	Calibration is timestamped and re-run each session. A side outside the ceiling is marked unusable.

4.2 Runtime state machine

```
STARTING ──(frames fresh ∧ ToF ok ∧ cue ack)──▶ READY
READY ──(ToF invalid ∨ low light ∨ confidence low)──▶ DEGRADED   # announce once; only supported claims
READY/DEGRADED ──(frame age > 500 ms ∨ no heartbeat > 500 ms from either board ∨ cue NACK ∨ phone
battery critical)──▶ UNAVAILABLE
UNAVAILABLE: play "system off" on any surviving cue output + phone speaker/vibration; suppress all
obstacle cues; retry link
any ──(user pause)──▶ PAUSED   # explicit, spoken confirmation; never automatic
```

The 500 ms frame-age and per-board heartbeat timeouts are **PROPOSED**; heartbeat interval targets 100 ms. They leave time inside the untested ≤ 1 s fail-loud goal and must be tuned at Gate G7. A bank without telemetry cannot report low charge before its outputs die.

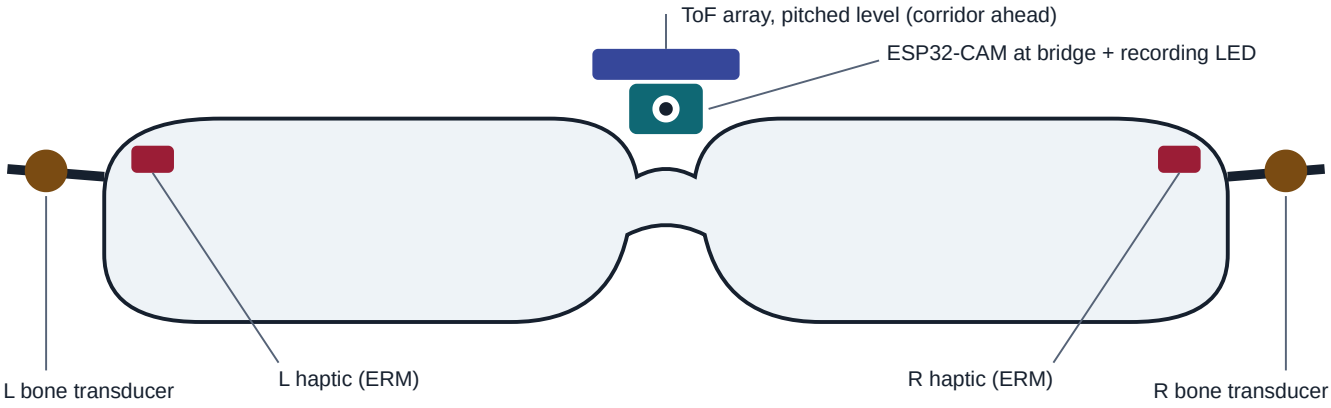
4.3 Wire contract (v0)

```
ESP32-CAM → phone (UDP/TCP over hotspot; little-endian; versioned)
  FRAME   {ver:u8, seq:u32, t_cap_us:u64, w:u16, h:u16, len:u32, jpeg[len]}
  BEAT    {ver, seq, t_us, faults:u16}
XIAO ESP32-C3 → phone
  RANGE   {ver, seq, t_us, mm:u16[64], status:u8[64]}
  BEAT    {ver, seq, t_us, faults:u16}
Phone ↔ each board (clock estimation; the camera board accepts no cue commands)
  PING    {t_phone_us}  →  PONG {t_phone_us, t_board_us}
Phone → XIAO ESP32-C3
  CUE     {ver, seq, t_issued_phone_us:u64, cue_id:u8, side:{L,R,BOTH},
           gain_l_db:i8, gain_r_db:i8, haptic:{pattern:u8, l:u8, r:u8}, ttl_ms:u16}
  PROFILE {ver, ceiling_db:i8, offsets_db[bands][2]}
```

There is no board-to-board bus: the phone is the only integrator. The phone estimates a separate clock offset for each board, timestamps receipt, and discards `FRAME` or `RANGE` whose offset-corrected capture time or receive age exceeds the freshness bound. The cue board enforces the gain ceiling independently. It compares `t_issued_phone_us + ttl_ms` against its offset-corrected current time, dropping a delayed `CUE` rather than resetting its lifetime on arrival. The C3's watchdog plays its system-off cue when phone PINGs stop while the board is powered; the phone uses its own speaker/vibration if a board disappears. Clock uncertainty, command expiry and both failure directions must be tested in Gates G6–G7. Power-bank battery percentage is not reported by either board unless external sensing is added; loss of its power is detected by missing heartbeats on the phone. **DESIGN**

CHAPTER 5

Wearable hardware



Cue board, amps and drivers sit behind the ear or on a rear strap; the ESP32-CAM mounts at the bridge. Two boards, separate power runs.

Figure 5.1. v0 layout. Weight is kept off the nose. The lenses are interchangeable (clear or tinted) and do not affect sensor alignment.

5.1 Sensing options

Configuration	What it gives	Cost and risk	v0?
Mono RGB only	Object classes and bearing (left/centre/right) from image position	No distance for arbitrary obstacles. Common detectors cover objects such as people, chairs and cars, but not branches or poles INFERENCE	Fallback
Mono RGB + 8×8 ToF	Class-agnostic distance in a forward corridor, plus semantics from the camera. VL53L5CX module: 4 m, 63° diagonal FOV, 15 Hz at 8×8, Class 1 940 nm laser [20]	$25.95 [20]. Loads ~90 KB of firmware over I²C at every boot [20]. Behaviour in sunlight and against glass is unmeasured	**Yes**
Dual RGB stereo	Dense depth after calibration	Needs synchronised exposure and stereo rectification. The ESP32-CAM carries one sensor and the C3 has no camera interface at all [13][59], so this needs two camera MCUs or a stereo module	No
Mono + learned depth	Relative depth. Depth Anything V2-Small is Apache-2.0, while its larger variants are non-commercial [23]	No metric scale unless anchored to the ToF. Phone compute load	Stretch
Stereo depth module (e.g. OAK-D Lite)	On-device depth	Expensive, multi-watt, harder on Android (secondary source only)	No
Meta Ray-Ban via DAT	Glasses camera stream up to 720×1280 at 2–30 fps over Bluetooth [6][58]	No ToF or haptics. Latency undocumented. Requires owning the glasses. Gen 3 not listed [6]	Optional adapter

Why ToF instead of a second camera? Head-level hazards such as branches, sign edges and open cabinet doors are rarely detector classes. The hazard decision must not depend on recognising the object, and a ToF array measures distance regardless of what the object is. The camera adds meaning ("person", "door") and sharper bearing. DESIGN

5.2 Wearable board choice (two-board revision, 24 Oct 2026)

The XIAO ESP32-S3 Sense is not obtainable for this build, so the wearable splits into two boards that share no bus: a **cue board** (ToF over I²C, I²S stereo playback, two haptic outputs) and a separate **camera board** that only streams JPEG. Both join the phone's hotspot as independent clients (Chapter 4). DESIGN

Board	Relevant facts	Verdict
Seeed XIAO ESP32-C3 (3-pack)	RISC-V single core, 400 KB SRAM and 4 MB flash, Wi-Fi b/g/n + BLE 5, exactly one I²C and one SPI bus, 11 GPIO, 21×17.8 mm, no integrated camera [59]. Vendor-typical 75 mA with Wi-Fi active [59]	**Selected: cue board.** Micro Center 3-pack; product page blocked (HTTP 403), so price and stock are unverified — confirm in person. One board used, two spares
ESP32-CAM (classic ESP32)	The esp32-camera driver lists ESP32, S2 and S3 as supported SoCs [13]. Practical streaming needs PSRAM: the driver requires it for anything above CIF-or-lower JPEG [13]	**Selected: camera board.** Candidate purchase: Inland pair at Micro Center, price and stock unverified (HTTP 403). Camera test gated on the actual module's PSRAM and sensor (Gate G0)
Seeed XIAO ESP32-S3 Sense	8 MB PSRAM, camera board (OV3660 or OV2640 since 2025-06-30), microphone, one DVP camera interface [12] [14]	**Not selected:** unavailable to this build. Documented single-board alternative for a later revision ($13.99 online when checked 24 Oct [14])
Raspberry Pi Zero 2 W	Quad A53, CSI-2, H.264 1080p30 encode, Wi-Fi and BT 4.2, 5 V/2.5 A supply spec [53]	Overkill for a "dumb" frame. Power and boot time. Out of stock at Adafruit
Raspberry Pi Pico 2 W	RP2350, 520 KB SRAM, no camera port [54]	Cannot carry the camera role
ESP32-P4	USB 2.0 high-speed, MIPI-CSI with ISP, H.264 up to 1080p30, but no radio [55]	Roadmap: wired high-speed or Wi-Fi via a companion chip

5.3 Output hardware

- **Stereo bone conduction, played on the glasses.** Two MAX98357A I²S amplifiers ($5.95 each [36]) share one I²S bus. Each amp's channel select is configured so one plays left and one plays right, because the default setting mixes both channels. Each amp drives a bone transducer. Adafruit's 8 Ω / 1 W transducer is $8.95 but was *out of stock* on 17 Oct [36]. For prolonged skin contact, Adafruit recommends covering the transducer and its wire connections (for example with Sugru) so that current cannot pass through sweat [36]. Apply that rule to any substitute transducer. FACT → DESIGN
- **Haptics.** Two ERM disc motors ($1.95 each [36]). Drive them either from two DRV2605L haptic drivers ($7.95 each [36]) for rich waveforms, or from a logic-level MOSFET per motor with PWM for lower cost. The DRV2605L uses a fixed I²C address and the C3 has a single hardware I²C bus [59], so two of them need an I²C multiplexer; the MOSFET-PWM alternative avoids the problem entirely. INFERENCE (confirm against the TI datasheet at build time.)
- **Fallback audio.** A commercial bone-conduction headset paired to the phone, such as Shokz OpenMove at a $79.95 list price (variants showed sold out) or OpenRun at $129.95 [37]. Classic Bluetooth adds latency (§6.2), so in this mode haptics carry the urgent cue.

5.4 Power

The revision puts two Wi-Fi radios on the frame. Seeed lists 75 mA typical for the C3 with Wi-Fi active [59]; the ESP32-CAM's draw is unmeasured on the actual module, and both boards' transmit peaks — plus amplifier and motor bursts — are what the supply must survive. v0 uses a pocket USB power bank feeding both boards over separate runs, which keeps the battery off the head and avoids building a Li-Po charging circuit during a hackathon.

DESIGN This revision makes no one-cable or peak-current claim: Gate G0 checks the two-board distribution, and Gate G11 measures runtime and thermal.

GPIO budget: the C3 has one hardware I²C bus [59], shared by the ToF array and, if DRV2605L drivers are fitted, by both of them behind an I²C multiplexer — there is no second bus to claim. I²S (3 pins) plus motor PWM (2 pins) must fit the remaining header pins, and the pin map must be fixed before soldering. GATE G0

CHAPTER 6

Links: Wi-Fi, USB, BLE, LE Audio, Auracast

6.1 Camera and sensor transport

Link	Facts	Use in Determine
Wi-Fi to the phone hotspot	Both wearable boards join the hotspot as stations: the ESP32-CAM streams JPEG and the C3 sends ToF grids and acknowledgements. esp32-camera recommends JPEG, because YUV and RGB strain PSRAM bandwidth when Wi-Fi is on [13]. Both boards and the phone share one hotspot channel's airtime	**Primary.** QVGA–VGA JPEG at 10–15 fps target. Joining the phone's hotspot keeps the phone's cellular internet, which joining a glasses-hosted access point would not. INFERENCE Effective throughput with two stations is unmeasured; Gate G1 measures it
USB-C UVC	Android supports UVC webcams through Camera2, but only if the device maker enables the external camera provider. AOSP says it "isn't designed" for high-speed streaming or AR, and its sample configuration bounds 640×480 at 30 fps [10]	Secondary, only after a per-phone check with a UVC camera. A user-space UVC library is a further fallback
BLE GATT	ESP32-to-ESP32 throughput is about 0.73 Mbit/s on the 1M PHY and about 1.4 Mbit/s on the 2M PHY. Phone throughput is lower and undocumented [15]	Provisioning, health, and possibly cues. Not video
Meta DAT	The camera stream degrades resolution first, then frame rate, and does not drop below 15 fps. A2DP and HFP audio are mutually exclusive [6][7]	Alternative FrameSource for teams that own supported glasses

6.2 Audio paths and latency

Path	Latency facts	Role
Phone → Classic BT headset (A2DP)	Typically 100–200 ms [16]	Fallback and spoken detail only
Phone → LE Audio unicast (LC3)	LC3 uses 10 ms frames. Best case is "just over 20 ms" end to end. Every receiver must support a 40 ms presentation delay, and hearing-aid live use stays at ≤ 20 ms [16]. Android has had LE Audio since Android 13 [19]	Roadmap: deliver cues to LE Audio hearing aids and earbuds
Auracast broadcast	One-to-many, connectionless. The transmitter never knows who is listening [17]. Android support is limited to some devices (Pixel 8 and later, Galaxy S23 and later, some Xiaomi) [18]	Demo extra: judges listen on their own Auracast earbuds through an encrypted stream. Never the personal safety path
Phone → cue board cue ID (UDP) → stored earcon	LAN hop plus I²S playback. Unmeasured	**Primary urgent path.** True stereo, per-side gain applied on the cue board

> **Why the glasses play the earcons.** The "dumb glasses" principle is about where decisions are made, not about refusing to store a 2-second sound bank. The phone decides *which* cue to play and *how loud* each side is. The glasses play it, which removes 100–200 ms of A2DP latency and the cost of a headset. DESIGN

6.3 End-to-end latency budget

Stage	Budget	Basis
Exposure + on-chip JPEG encode	≤ 60 ms	PROPOSED
Wi-Fi transfer (~30 KB frame)	≤ 40 ms	~1 MB/s effective throughput assumed for one station; the C3 shares the hotspot's airtime, so this is unmeasured for two producers. INFERENCE
Decode + detector + fusion on phone	≤ 80 ms	Benchmark the actual model on the actual phone
Policy + cue dispatch + playback start	≤ 40 ms	UDP on LAN. INFERENCE
Capture-to-cue p95 target	**≤ 250 ms**	**Target kept from the single-board design; unmeasured with two Wi-Fi producers. The ToF path skips the camera stages and should be faster**

At a walking speed of 1.4 m/s, every 100 ms of latency is about 14 cm of travel. INFERENCE So 250 ms is about 35 cm before human reaction time is even counted. Warning distances in the hazard policy have to include both.

CHAPTER 7

Perception and hazard policy

7.1 Pipeline

1. **Ingest and freshness gate.** Two independent producers: the ESP32-CAM's frames and the cue board's ToF grids each carry their board's clock, a per-board offset estimated by PING/PONG, and the phone's receive timestamp. Drop any input whose offset-corrected capture time or receive age exceeds the freshness bound. Every stage records a timestamp, for Gate G6.
2. **Range corridor (class-agnostic).** Split the 8×8 ToF grid into a *corridor* (central columns, head-to-chest rows) and left and right flanks. Use the minimum valid distance per region. Invalid zones count as UNKNOWN.
3. **Semantics.** MediaPipe Object Detector in `LIVE_STREAM` mode with a category allowlist and a score threshold [21], running on the LiteRT runtime [22]. Classes are used for *labels* ("person", "door", "chair") and for bearing, never as the only trigger for a near-field hazard.
4. **Fusion.** Match detections to ToF regions by bearing. If a ToF return lies in the corridor and no detection matches, report it as a generic "obstacle".
5. **Proximity band.** FAR / MID / NEAR come from ToF distance when it is valid. Without ToF, use only *relative* cues such as bounding-box growth over time, and never speak metres.
6. **Policy.** Pick at most one alert, with persistence and hysteresis (below).

7.2 Hazard policy (v0 rules)

```
inputs: corridor_min_mm, left_min_mm, right_min_mm (valid flags), detections, head_yaw_rate
band(d): NEAR if d < 1000; MID if d < 2000; FAR if d < 3000; else NONE      # PROPOSED mm
enter alert:  same region ∧ band ≥ FAR for ≥ 3 consecutive ToF samples (~200 ms @15 Hz)
exit alert:   band NONE for ≥ 5 samples, or region changes (re-enter with new cue)
priority:     NEAR corridor > NEAR flank > MID corridor > MID flank > FAR corridor
rate limit:   ≤ 1 alert onset / 700 ms; repeat cadence set by band (Chapter 8)
suppress:     head_yaw_rate high (scanning) → hold FAR cues; NEAR always passes
label:        matched detection class if confident, else "obstacle"
never:        output CLEAR; speak distance without valid ToF; let description pre-empt NEAR
```

The bands are PROPOSED. With 250 ms of system latency and about 1 s of human reaction at 1.4 m/s, a hazard reported at 1.0 m leaves very little margin. That is why the FAR and MID bands exist, and why the NEAR band is a last-moment warning, not the main mechanism. INFERENCE White-cane mode MAY suppress low-zone returns that the cane already covers. The Ara wearable uses a similar mode [1].

7.3 Optional scene description (Describe loop)

The user asks with a long press or voice: "What's ahead?" — never automatically. If the user has opted into cloud description and the service is available, the phone redacts faces and sensitive detail before upload; imperfect redaction is a privacy risk, so upload happens only on the user's explicit choice. A small arm64 Lambda API on Graviton2 [65] holds the prompt — two sentences, surroundings first, no guesses about people — and a small vision-language model describes the image: SmolVLM via Hugging Face Optimum Neuron on one Inf2 instance [62] [63]. The model family is documented for Inf2, but this deployment is **untested by us**. Trainium2 is the documented alternative [60][61][64], and Bedrock Nova Lite is a fallback that does not guarantee the silicon [66]; new AWS accounts start with zero Inf/Trn quota, so accelerator access must be requested early [67]. If no description service is ready, the phone says description is unavailable; it does not improvise an untested local model. A returned answer is prefixed "may be wrong". MID or NEAR hazards interrupt it, and it is inactive while the system is UNAVAILABLE. Response time is unmeasured; the design plans for seconds, not milliseconds. DESIGN

7.4 "Cannot see" list (spoken during onboarding)

Glass doors and windows · thin wires and ropes · drop-offs, curbs and stairs going down · objects outside the forward field of view · fast-moving vehicles · glare, darkness, rain and fog · anything beyond about 3 m.

CHAPTER 8

The cue language

The requirement is a sound that is granular, easy to identify and level-aware. Determine treats the cue as a small grammar with four independent dimensions, so that no single dimension, and especially not loudness alone, has to carry everything. Brewster's guidelines warn against identification by register or intensity alone [24]. FACT → DESIGN

Dimension	Encoded by	Values	Rationale
What (family)	Timbre + rhythm motif	Obstacle: short wood-block double. Person: two-tone marimba. System: descending three-note. Ready: single soft tick, once	Brewster: timbre and rhythm are the strongest discriminators [24]
Where (side)	Calibrated side + matching haptic + optional spearcon	Left · Centre (both sides together, both motors) · Right	Redundant, so it survives single-sided deafness and the skull-crossing of bone-conducted sound [26][28]
How close	Repetition rate (parking-sensor metaphor)	FAR 1 per s · MID 2.5 per s · NEAR 5 per s plus urgent motif	Matches the car-sensor mental model. Stays within about 6 notes per second [24]
How urgent	Limited level step + haptic intensity	Base at calibrated sensation level. NEAR at +6 dB *within the ceiling*	Level supports urgency but is never the only signal. Stays 10–20 dB above threshold [24]

8.1 Timing grammar

```
note ≥ 83 ms (motif notes)   gap between serial cues ≥ 100 ms   max 6 notes/s      [24]
pitch range 300 Hz – 4 kHz  (inside Brewster 125 Hz–5 kHz and inside transducer response)
FAR:  ■■ ............ ■■ ............  (1/s)
MID:  ■■ ... ■■ ... ■■ ...            (2.5/s)
NEAR: ■■■ ■■■ ■■■ + haptic buzz       (5/s, urgent motif, +6 dB within ceiling)
SYSTEM OFF: ■ □ □ descending, repeated every 5 s until resolved, plus phone vibration
```

The lower edge of the pitch range assumes the bone transducer's 300 Hz–19 kHz response [36]. Recheck it against whatever transducer is actually fitted.

8.2 Spoken layer

Spearcons are sped-up words such as "left", "door" or "person". They are optional and follow the earcon, never replace it. They are on by default during training and become opt-in once the wearer is fluent. Clock-face phrasing ("11 o'clock, 2 metres") is used only when ToF is valid, following the study finding that precise phrasing was preferred [2][25]. DESIGN

8.3 Haptic mapping

Event	Left motor	Right motor
Obstacle left / right	Pulse at band rate / off	Off / pulse at band rate
Obstacle centre	Both pulse together at band rate	

Event	Left motor	Right motor
NEAR (any side)	Strong 150 ms buzz on the hazard side(s), then band pulses	
System unavailable	Long-short-short on both sides, plus phone vibration	

Haptics are the *primary* directional channel in single-sided-deafness mode, and a confirming channel otherwise. The BuzzClip result warns that weak, frequent vibration can be confused with cane-tip feedback [2], so intensity and cadence are user-tunable and validated in Gate G9. DESIGN

8.4 Cue bank as data

```
cue_bank.json  (shipped to glasses flash; phone references by cue_id)
{ "obstacle_far":  {"wav":"ob2.wav", "period_ms":1000, "haptic":"pulse40"},
  "obstacle_mid":  {"wav":"ob2.wav", "period_ms":400,  "haptic":"pulse60"},
  "obstacle_near": {"wav":"ob3u.wav","period_ms":200,  "haptic":"buzz150","level_step_db":6},
  "person_*":      {"wav":"pe2.wav", ...},
  "system_off":    {"wav":"sys3d.wav","period_ms":5000,"haptic":"lss","both":true} }
```

CHAPTER 9

Calibration for asymmetric hearing

Goal. Both sides of the cue should be *equally perceptible* to this wearer, with this frame position and this output device, without ever exceeding a hard ceiling. This is perceptual matching. It is not a hearing test or a diagnosis, and it does not restore hearing. DESIGN



Figure 9.1. Calibration flow, roughly 2–3 minutes and fully operable with TalkBack and screen-reader gestures.

1. **Fit and device.** The app asks the wearer to seat the frame. It records which output device is in use (glasses transducers or a headset) and the wearer's self-reported weaker side. It warns about ambient noise if the phone microphone reads high.
2. **Per-side detection threshold.** An adaptive 2-down-1-up staircase, which converges near 70.7% correct [57], starts from a quiet level. The wearer answers "heard / not heard" by tapping. It runs in three bands that match the cue timbres (for example 500 Hz, 1 kHz and 2 kHz) and stops after 6–8 reversals. PROPOSED
3. **Presentation level.** Per band and side, the level is threshold plus a sensation level chosen by the wearer between 10 and 20 dB [24], clamped to the ceiling.
4. **Loudness balance.** Alternate the real "obstacle" cue left and right. The wearer nudges a balance control until both sides feel equally loud, which corrects for cue spectrum and for the bone crossing the skull. Repeat once to check consistency.
5. **Verify and choose a mode.** Play five random-side cues and have the wearer name the side.
   - **Stereo mode:** at least 4 of 5 correct and no side at the ceiling.
   - **SSD mode:** one side cannot be detected at the ceiling, or side naming is at chance. Audio becomes identical on both sides (mono). Direction moves to haptics plus spearcons.
   - **Haptic-only mode:** for wearers who prefer it or cannot use the audio.

9.1 Profile and safety rules

```
HearingProfile {
  device_id, created_at, fit_note,
  mode: STEREO | SSD | HAPTIC_ONLY,
  ceiling_db,                      # from device characterisation; glasses enforce independently
  bands: [{hz, thr_db_L, thr_db_R, sl_db, offset_db_L, offset_db_R}],
  haptic: {gain_L, gain_R, pattern_set},
  verified: {trials:5, correct:n}
}
```

- **Ceiling first.** No calibration step, slider or urgency step can exceed `ceiling_db`. If the weak side would need more, it is marked unusable. The app does not push more energy into it. DESIGN
- **Re-run every session**, because how the frame couples to the head changes each time it is worn. A 20-second quick check suffices if the previous profile is under 7 days old. PROPOSED

- **Dose.** Track cumulative cue-on time. Cues are short and sparse by design. Until the transducers are measured on an artificial mastoid, Determine makes no claim of equivalence to the WHO and ITU air-conduction limits [4] [5].
- **Professional care.** The app recommends an audiologist for anyone with known hearing loss and never displays results as audiometry.
- **OS features stay reachable.** Android mono audio and balance settings remain available. Determine's profile works alongside them, not as a replacement.

CHAPTER 10

Safety, privacy and regulatory boundary

10.1 Threat and hazard ledger

Hazard	Mitigation	Residual risk
Missed obstacle (glass, wire, out of FOV)	"Cannot see" list. Secondary-aid positioning. Cane mode	Inherent. Disclosed
False reassurance	No "clear" cue exists. UNAVAILABLE state. TTL on cues	Users may over-trust silence. Address it in training
Over-alerting and fatigue	One alert at a time, rate limit, persistence and hysteresis	Tune against measured false-alert rate
Masking environmental sound	Short sparse cues, open ears, cue-on duty tracked [52]	Measure in Gate G10
Excess vibration or sound level	Ceiling enforced on both phone and glasses. Session timer	No bone-conduction dose standard available
Electrical and thermal	Pocket USB bank, no head-mounted Li-Po in v0, strain relief, temperature in heartbeat	Hackathon-grade wiring. Supervised use only
Bystander privacy	Visible recording indicator, no retention, demo signage and consent	Camera on face may still worry bystanders
Spoofed cue commands	v0 hotspot is WPA2 with a unique password. Roadmap: authenticated session key	LAN attacker during demo. Low impact, but logged

10.2 Regulatory boundary

21 CFR 886.5900 defines an *electronic vision aid* as a device that translates visual images of objects into tactile or auditory signals for a person with impaired vision or blindness. It is Class I and exempt from 510(k) premarket notification [31]. FACT Marketing Determine as a mobility aid would likely place it in this category. Being exempt from 510(k) does not mean being exempt from every FDA obligation, so get legal review of the remaining duties before any sale. The hackathon prototype is a research demonstration. INFERENCE

10.3 Human-subjects boundary

Testing with blind, low-vision or hearing-impaired participants beyond informal feedback SHOULD go through the university's human-subjects process, include an O&M specialist, and have informed consent and a spotter. Every HackTX demo uses a staged indoor course with a sighted spotter. DESIGN

10.4 Meta platform rules (if the DAT adapter is used)

Meta's acceptable-use policy prohibits tampering with the capture LED. It allows face or voice identification only for accessibility apps with Meta's prior approval [8]. Determine does no face identification. FACT

CHAPTER 11

HackTX 26 build plan

11.1 Event facts (official site)

- Hacking runs from **8 am Saturday 24 Oct to 5 pm Sunday 25 Oct 2026** on the UT Austin campus, fully in person. Teams have at most 4 people. The event is free and does not reimburse travel [32].
- Rule: **"All work must be done at the event. You can't demo something you didn't build."** Attendees follow the MLH Code of Conduct [32].
- Prizes exist for challenges, with details "revealed at the opening ceremony". No judging rubric, tracks or prize values are published [32].
- The listed sponsors include Meta (highlighted tile), Visa, Annapurna (linked to Annapurna Labs), NMC2, TFS, BHW, D. E. Shaw and Capital One [32].
- Conflict on the site: the application banner says "Apply by Sept 15" while the FAQ says Sept 11. Walk-in admission is first come, first served, is not guaranteed, and the site discourages non-Austin attendees from travelling for it [32]. **Confirm acceptance in the portal before travelling.**

> **Implication:** all work, including this document, is done at the event: buy parts, read documentation, plan, write firmware, write the app and assemble the frame there. Email `hello@freetailhackers.com` to confirm whether cue sound files and phone compatibility checks are allowed. Get the answer in writing.

11.2 Team roles (4 people)

Role	Owns
Firmware	ESP32-CAM JPEG stream; XIAO ESP32-C3 ToF, UDP protocol, I²S cue player, haptic driver, watchdog, heartbeat
Android	FrameSource and RangeSource ingest, MediaPipe detector, fusion, HazardPolicy, latency logging
Audio / UX	Cue bank, calibration flow, TalkBack-accessible UI, spearcons, judge diagnostics overlay
Hardware / demo	Frame mounting, wiring, power, the staged course, test protocol, pitch and Devpost

11.3 Timeline with abort branches

Window	Milestone	Abort or branch
Sat 8–12	Two-board bring-up: ESP32-CAM JPEG reaches the phone over the hotspot; C3 ToF grid streams; two-board power distribution checked; pin maps fixed; cue player plays a stored WAV on both sides	Camera module lacks PSRAM or has an unsupported sensor → drop resolution (CIF-or-lower JPEG needs no PSRAM [13]), else Tier 0 (phone camera on headband) while debugging. Transducers missing → phone-to-headset fallback
Sat 12–18	End-to-end loop: ToF corridor → policy → glasses cue. Detector labels added. Timestamps logged	Detector slow → ToF-only hazard with generic "obstacle" label
Sat 18–24	Calibration flow, three modes, haptics, UNAVAILABLE handling	DRV2605L trouble → MOSFET PWM
Sun 0–8	Gates G6–G8 on the staged course. Tune bands and rates. Freeze features	No new features after 8 am

Window	Milestone	Abort or branch
Sun 8–12	Optional adapters: Meta DAT through Mock Device Kit, Auracast demo broadcast (only on supported phones)	Skip if any core gate is failing
Sun 12–17	Pitch rehearsal, Devpost, demo resets, spare batteries	—

11.4 Three-minute judge demo

1. **Hook (20 s):** the parking-sensor line. Hold up the two-board frame next to the costs of NOA ($4,990) and OrCam ($4,250).
2. **Live course (75 s):** a sighted spotter walks the wearer past a head-height foam bar, a person, and a left-side then right-side obstacle. The judge diagnostics screen shows frame age, ToF grid, chosen alert and measured latency.
3. **Hearing (40 s):** switch the profile to SSD mode live. The urgent cue stays audible and direction moves to haptics plus spearcons. Show the ceiling stopping a forced gain increase.
4. **Failure (25 s):** unplug the camera. The system-off cue plays within the measured bound, and no "clear" is ever heard.
5. **Evidence (20 s):** the logged p50/p95 latency, the miss and false-alert table, and the BOM.

11.5 Sponsor alignment, without forcing it

Meta

Wearables DAT 1.0 (rollout from 30 Sep 2026) gives mobile apps camera, microphone, audio and IMU access on supported AI glasses. Its Mock Device Kit simulates the glasses camera without hardware [6]. Determine's `FrameSource` seam lets a DAT adapter feed the *same* hazard loop. The story: Determine runs on commodity dumb glasses *and* on Ray-Ban Meta. Claim it only if the adapter was built at the event.

Annapurna Labs (AWS)

Annapurna Labs is AWS's Austin-based chip group behind Nitro, Graviton, Inferentia and Trainium [33]. The honest connection is *system-first co-design*: Determine places computation in the frame, the phone or the cloud according to latency, power and failure budgets (Chapter 6), and measures them. The optional Describe loop targets that silicon: a Graviton2 Lambda API holds the prompt, and SmolVLM would run on Inf2 via Optimum Neuron, with Trainium2 the documented alternative (§7.3) [34][35][62]–[65]. They stay out of the hazard loop.

Winning overall is an ambition, not something this document can engineer. What it can engineer is the set of things judges reward: a complete working loop, a clear human need backed by evidence, measured numbers, honest limits and a memorable demo.

CHAPTER 12

Sourcing in Houston and bill of materials

12.1 Verified Houston-area stores

Store	Address	Hours / phone	What to look for
Micro Center Houston	5305 S Rice Ave, Houston, TX 77081	Mon–Sat 10–9, Sun 11–6 (chain hours) [38]	The XIAO ESP32-C3 3-pack and the ESP32-CAM pair (both online pages blocked — verify price and stock in person), USB-C cables and OTG adapters, power banks, possibly ToF and haptic breakouts
Electronic Parts Outlet (EPO)	3753-B Fondren Rd, Houston, TX 77063	Mon–Fri 9–6, Sat 9–5, Sun closed · (713) 784-0140 [39]	Says it stocks Arduino, SparkFun, Raspberry Pi, Pololu and Adafruit lines. Call about specific parts
Ace Electronics	3210 Antoine Dr, Houston, TX 77092	Mon–Fri 8–5 only · 713-688-8114 [40]	Vibration motors, wire, protoboard, heat shrink, MOSFETs
Best Buy Galleria	5133 Richmond Ave, Houston, TX 77056	(713) 552-1512 [41]	Commercial bone-conduction headset (fallback audio). Check stock online first
Micro Center Austin	12707 N Mopac Expy, Austin, TX 78727	Opens 10 am Saturday [38]	Emergency top-up during HackTX (hacking starts at 8 am)

No electronics supplier in Galveston was verified, so plan Houston runs. Live shelf stock could not be confirmed at any of these stores, because the retailer pages blocked automated reads. Ace's and Altex's online availability flags show web-store stock, not shelf stock. **Call before driving.**

12.2 Bill of materials, recommended v0 ("Dumb Frame")

Item	Qty	Unit	Total	Source and status
Seeed Studio XIAO ESP32-C3, 3-pack — **cue board** (ToF host, I²S playback, haptics)	1 pack (3 boards)	—	—	Selected Micro Center listing; page blocked (HTTP 403, 18 Oct) — price and local stock unverified, confirm in person. One board used, two spares
ESP32-CAM module pair (Inland) — **camera board**, candidate purchase	1 pair (2 modules)	—	—	Candidate Micro Center listing; page blocked (HTTP 403, 18 Oct) — price and local stock unverified. Camera use gated on the actual module's PSRAM and sensor (Gate G0) [13]
SparkFun VL53L5CX 8×8 ToF (Qwiic Mini)	1	$25.95	$25.95	SparkFun, listed in stock when rechecked 18 Oct [20]
MAX98357A I²S amp (L and R)	2	$5.95	$11.90	Adafruit, in stock [36]
Bone conductor transducer 8 Ω 1 W	2	$8.95	$17.90	Adafruit, **out of stock** when rechecked 18 Oct [36]. Find an equivalent bone transducer and insulate it before skin contact

Item	Qty	Unit	Total	Source and status
DRV2605L haptic driver	2	$7.95	$15.90	Adafruit, in stock [36]. Needs an I²C mux on the C3's single bus, or MOSFETs from Ace/EPO
I²C mux for two DRV2605Ls, or MOSFET-PWM drive instead	1 option	—	—	Choose and price at G0; the two identical-address drivers cannot coexist on the bare C3 bus
Vibrating mini motor disc	2	$1.95	$3.90	Adafruit, in stock [36]. Ace lists a $5 vibration motor
Wire, protoboard, heat shrink	1	$12.80	$12.80	Ace online listing [40]
Glasses frame (clear or tinted), mounting tape, Velcro	1	—	$10.00	Allowance, price not verified
USB power bank with two outputs, two compatible power leads; camera programming adapter if needed	1 set	—	—	Price and verify both output ports under simultaneous Wi-Fi load; allow for ESP32-CAM programming hardware
Subtotal of priced line items			**$88.35**	**Uses the two DRV2605L prices and includes out-of-stock transducers. Excludes both board packs, mux, power/leads, programming adapter and $10 provisional frame allowance. Not a purchasable grand total**

12.3 Tiers

Tier	Contents	Cost status	Use
Tier 0: phone-only	Android phone on a headband using its own camera, plus available headphones and phone vibration. Same app, using the PhoneCamera FrameSource	Unquoted	Software proof; not the glasses
Tier 1: minimum frame	C3 cue board + ESP32-CAM + ToF + 2 motors (MOSFET drive) + supplies + frame + power. Audio from phone to existing headphones	Unquoted	Camera and haptic demo
Tier 2: recommended	Full table above: two boards, ToF, stereo bone cues played on the cue board, haptics	Incomplete subtotal	Full demo; transducer substitute needed
Fallback audio	Replace DIY transducers and amps with a bone-conduction headset	Price at purchase	Haptics carry urgency

Other listed products (24 Oct 2026): Ray-Ban Meta Gen 3 from $449 [42], Envision Ally Solos $699 [43], OrCam MyEye 3 Pro $4,250 [44], NOA $4,990 [46], .lumen €9,999 [47]. They do different jobs. Determine's two-board total is unquoted, so no like-for-like cost-of-entry comparison is claimed.

12.4 Purchase timeline (two-board revision, Saturday 24 Oct 2026)

1. **Sat 24 Oct:** At Micro Center Austin (opens 10 am), confirm both board prices and stock; check the camera module's PSRAM and sensor [13]. Buy two power leads, a two-output bank and a programming adapter if needed. Pick up ToF and cue parts in the same trip. Check a bone-headset fallback.
2. **Sat 24 Oct (afternoon):** Call EPO (9–6) about transducers and sensor substitutes; Ace (8–5) for motors, MOSFETs and wiring; buy local equivalents if anything is out.
3. **Sun 25 Oct:** Micro Center Austin opens at 10 am for anything still missing.

CHAPTER 13

Verification gates and failure catalogue

Gate	Test	Pass criterion (PROPOSED unless stated)
G0	Bring-up on two boards: ESP32-CAM JPEG frames and cue-board ToF at 15 Hz, with both pin maps and the two-board power distribution fixed. Record the camera module's actual PSRAM and sensor	10 minutes, no resets, no corrupt JPEGs, no brown-outs when both radios transmit
G1	Both Wi-Fi producers stream to the team's phone over one hotspot simultaneously	≥ 10 fps QVGA/VGA JPEG sustained 10 min with the ToF stream active. Per-board frame and range loss plus staleness logged
G2	USB UVC on the team's phone (optional path)	Enumerates via Camera2 or a user-space library. Otherwise mark it unsupported
G3	Cue path: phone command → glasses earcon, both sides	True L/R separation confirmed. p95 command-to-sound measured
G4	Gain ceiling enforced on the glasses	A forged CUE above the ceiling is clamped
G5	Calibration repeatability	Two runs within ±3 dB per side, same fit
G6	End-to-end latency on the course	p50/p95/max over ≥ 100 events. Target p95 ≤ 250 ms (report the actual value either way)
G7	Fault injection: unplug or power down either board, kill app, Wi-Fi off, low battery	Unavailable cue ≤ 1 s. No stale cue after the fault
G8	Detection on the staged course: head-height bar, person, left and right obstacles, 10 repeats each	Miss and false-alert table published. No "works" claim without it
G9	Cue identification after 5 minutes of training	About 90% correct family and side, in ambient noise (the number is arbitrary and needs a pilot)
G10	Masking check: localise real sounds with cues on versus off	Report the difference. Reduce cue duty if it is large [52]
G11	Thermal and battery over 30 minutes	Frame touch temperature comfortable. Runtime measured

13.1 Failure catalogue

Failure	Detection	Response
Phone lacks external-camera HAL	G2	Wi-Fi path (default anyway)
ESP32-CAM module has no PSRAM or an unsupported sensor	G0 bring-up; module markings vs the esp32-camera lists [13]	The driver requires PSRAM above CIF-or-lower JPEG [13]: drop resolution, else Tier 0 phone camera
PSRAM contention corrupts frames under Wi-Fi [13]	JPEG decode errors	Lower resolution or frame rate. Use JPEG only
Hotspot drops or the phone roams to another network	Heartbeat timeout on both boards	UNAVAILABLE, then auto-rejoin. Pin the SSID
One board drops off the hotspot while the other stays up	That board's BEAT times out	Fail loud: UNAVAILABLE, never a half-working "all clear". Auto-rejoin

Failure	Detection	Response
Clock-offset estimate drifts; a stale frame or grid arrives	Freshness gate (offset-corrected age + receive age)	Discard the input, re-estimate from PING/PONG. Persistent staleness → UNAVAILABLE
ToF null zones in sunlight or on glass	Status flags	DEGRADED announcement. Camera-only labels
Detector misses an unusual obstacle	G8 log	The ToF corridor still triggers "obstacle"
A2DP latency in fallback audio [16]	G3 measurement	Haptics carry urgency. Audio carries labels
Headset multipoint switches to another device	Audio route change callback	Alert on phone. Disable multipoint for the demo
Wearer cannot lateralise bone cues	Calibration verify step	SSD mode: mono audio, haptic and spoken direction
Brown-out when both boards transmit at once	Reset reason in heartbeat	Bulk capacitance, short separate power runs, power bank with adequate output

CHAPTER 14

Roadmap after HackTX

Track	Next step	Gate before claiming
User research	Co-design sessions with blind and low-vision travellers, O&M specialists, and people with unilateral hearing loss	Ethics approval. No blindfolded substitutes [2]
Hearing-aid output	Send cues directly to LE Audio hearing aids and earbuds (unicast, LC3), so wearers who already use hearing aids need no extra transducer [16][19]	Device support plus measured latency
Auracast	Encrypted broadcast so an O&M instructor or companion can monitor cues in training [17][18]	Supported phones only. Never the safety path
Meta DAT	Production DAT adapter on supported Ray-Ban and Oakley Meta models, with bone and haptic output kept on a separate accessory [6][7]	Measured stream latency (undocumented by Meta)
Sensing	Synchronised stereo or ESP32-P4 with high-speed USB. A downward-pitched sensor for drop-off *research*	Per-class recall data with O&M review
Hardware	Custom flex PCB in the temple arms, integrated battery with protection, IP rating	Electrical, battery and RF compliance testing
Single-board wearable	Re-combine camera and cues on one SoC when a camera-capable board (e.g. XIAO ESP32-S3 Sense [14]) is reliably obtainable; removes the second radio and the per-board clock alignment	Availability plus a G0-equivalent two-function bring-up
iOS	Wi-Fi FrameSource. Apple documents UVC external cameras for iPad [56], so treat iPhone USB cameras as unsupported until tested	Parity of gates G1–G7
Regulatory	Classification review under 21 CFR 886.5900 [31]	Legal counsel

CHAPTER 15

Architecture decision records

ADR	Decision	Alternatives rejected	Revisit when
001	Phone is the only decision-maker	On-glasses inference (cost, heat, battery). Cloud inference (latency, offline failure)	A frame SoC can run the policy at similar cost and power
002	Mono RGB + 8×8 ToF for v0	Dual RGB stereo (sync and calibration). Ultrasonic (wide beam, weak bearing)	Synchronised stereo is affordable and measured outdoors
003	Wi-Fi hotspot transport for frames	BLE video (bandwidth). USB UVC by default (support varies by phone maker)	Gate G2 passes on the target phone fleet
004	Urgent cues played on the glasses from stored earcons	A2DP streaming (100–200 ms). Phone speaker (no direction, public)	LE Audio unicast is widely available
005	Direction is always redundant; no panning-only cues	HRTF spatial audio alone (4 of 13 struggled [2]; SSD cannot use it)	Never for SSD mode
006	Calibration matches perceived level per session, with a hard ceiling enforced twice	Fixed balance slider. Unlimited boost of the weak side	A transducer dose standard exists
007	No "clear" cue; silence means no report from a healthy system	Periodic "all clear" reassurance	Never
008	Cloud and generative description out of the hazard loop	LLM-driven navigation (22.2% wrong answers [30])	Never for hazard cues
009	Meta DAT is a FrameSource adapter, not a dependency	DAT-only build (cost, no haptics or bone output)	The team owns supported glasses
010	Android first	iOS first (no UVC, Classic audio only)	After gates G1–G7 pass on Android
011	Two Wi-Fi producers: XIAO ESP32-C3 cue board + separate ESP32-CAM camera board, no board-to-board bus (24 Oct 2026)	Single XIAO ESP32-S3 Sense (not obtainable for this build). Hardwired board-to-board bus (unverified wiring, no spare bus on the C3, new failure modes)	A camera-capable single board is reliably purchasable, or two-station hotspot throughput fails Gate G1

CHAPTER 16

Source register

Retrieved 24 Oct 2026, except [13], [20], [36], [59] and the AWS entries re-read for the 24 Oct two-board revision. Only the SparkFun ToF and Adafruit bone transducer price/stock were rechecked among the older-priced parts. Vendor pages state vendor claims. Product prices drift.

1. Lighthouse Guild, "The White Cane and Beyond" (2024). https://lighthouseguild.org/news/the-white-cane-and-beyond/
2. Pittet et al., secondary ETA evaluation with BVI participants, *Scientific Reports* (2026). https://pmc.ncbi.nlm.nih.gov/articles/PMC12909938/
3. Manduchi & Kurniawan, mobility-related accidents of blind travellers (2011). https://users.soe.ucsc.edu/~manduchi/papers/MobilityAccidents.pdf
4. WHO, "Deafness and hearing loss: Safe listening" (6 Mar 2026). https://www.who.int/news-room/questions-and-answers/item/deafness-and-hearing-loss-safe-listening
5. WHO-ITU, Global standard for safe listening devices and systems. https://www.who.int/publications/i/item/9789241515276
6. Meta, Wearables developer FAQ (DAT 1.0, supported devices, Mock Device Kit). https://developers.meta.com/wearables/faq/
7. Meta, DAT microphones and speakers. https://wearables.developer.meta.com/docs/develop/dat/microphones-and-speakers/
8. Meta, Wearables acceptable-use policy. https://wearables.developer.meta.com/acceptable-use-policy
9. AFB AccessWorld, Meta glasses review (Fall 2025). https://afb.org/aw/fall2025/meta-glasses-review
10. AOSP, External USB cameras. https://source.android.com/docs/core/camera/external-usb-cameras
11. Android Developers, USB host. https://developer.android.com/develop/connectivity/usb/host
12. Espressif, ESP32-S3 datasheet. https://documentation.espressif.com/esp32-s3_datasheet_en.pdf
13. Espressif, esp32-camera. https://github.com/espressif/esp32-camera
14. Seeed Studio, XIAO ESP32S3 Sense. https://www.seeedstudio.com/XIAO-ESP32S3-Sense-p-5639.html
15. Espressif FAQ, BLE throughput. https://docs.espressif.com/projects/esp-faq/en/latest/software-framework/bt/ble.html
16. Bluetooth SIG, *Introducing Bluetooth LE Audio*. https://www.bluetooth.com/wp-content/uploads/2022/01/Introducing-Bluetooth-LE-Audio-book.pdf
17. Bluetooth SIG, How to build an Auracast transmitter (rev. 16 Sep 2026). https://www.bluetooth.com/wp-content/uploads/2024/05/2403_How_To_Auracast_Transmitter.pdf
18. Google, LE Audio and Auracast support on Android (2025). https://blog.google/products-and-platforms/platforms/android/le-audio-auracast-support/
19. Android Developers, BLE Audio overview. https://developer.android.com/develop/connectivity/bluetooth/ble-audio/overview
20. SparkFun, Qwiic Mini ToF Imager VL53L5CX. https://www.sparkfun.com/sparkfun-qwiic-mini-tof-imager-vl53l5cx.html
21. Google AI Edge, MediaPipe Object Detector for Android. https://ai.google.dev/edge/mediapipe/solutions/vision/object_detector/android
22. Google AI Edge, LiteRT for Android. https://ai.google.dev/edge/litert/android
23. Depth Anything V2 (licences per model size). https://github.com/DepthAnything/Depth-Anything-V2
24. Brewster, Wright & Edwards, earcon design guidelines (1995). https://www.dcs.gla.ac.uk/~stephen/papers/HCI95.pdf
25. Walker et al., spearcons (2013). https://pubmed.ncbi.nlm.nih.gov/23516800/
26. Stenfelt, transcranial attenuation in unilateral deafness (2012). https://pubmed.ncbi.nlm.nih.gov/22193619/
27. Stanley & Walker, bone-conduction lateralisation (2006). https://sonify.psych.gatech.edu/publications/pdfs/2006HFES-StanleyWalker.pdf
28. Agterberg et al., CROS bone-conduction devices and localisation (2019). https://pubmed.ncbi.nlm.nih.gov/29703651/
29. Van Wanrooij & Van Opstal, monaural localisation (2004). https://pubmed.ncbi.nlm.nih.gov/15115811/
30. Gonzalez et al., multimodal LLM diary study (2026). https://arxiv.org/abs/2602.13469
31. 21 CFR 886.5900, Electronic vision aid. https://www.law.cornell.edu/cfr/text/21/886.5900
32. Freetail Hackers, HackTX 26 official site and its published bundle (FAQ, sponsors). https://hacktx.com/
33. About Amazon, inside Annapurna Labs. https://www.aboutamazon.com/news/aws/take-a-look-inside-the-lab-where-aws-makes-custom-chips
34. AWS, Inferentia. https://aws.amazon.com/ai/machine-learning/inferentia/
35. AWS, Neuron SDK. https://aws.amazon.com/ai/machine-learning/neuron/
36. Adafruit products: DRV2605L https://www.adafruit.com/product/2305 · motor disc https://www.adafruit.com/product/1201 · MAX98357A https://www.adafruit.com/product/3006 · bone transducer https://www.adafruit.com/product/1674
37. Shokz, OpenMove https://shokz.com/products/openmove · OpenRun https://shokz.com/products/openrun
38. Micro Center, store locations and hours. https://microcenter.zendesk.com/hc/en-us/articles/24790484266007-Micro-Center-Store-Locations-and-Addresses
39. Electronic Parts Outlet. https://epohouston.com/

40. Ace Electronics. https://ace4parts.com/
41. Best Buy Galleria store page. https://stores.bestbuy.com/tx/houston/5133-richmond-ave-291.html
42. Meta store, Ray-Ban Meta. https://www.meta.com/ai-glasses/ray-ban-meta/
43. Envision, Ally Solos shop. https://shop.letsenvision.com/products/ally-solos-glasses
44. OrCam MyEye 3 Pro (reseller price). https://www.getorcam.com/product-page/myeye-3-pro
45. WeWALK product. https://wewalk.io/en/product
46. NOA (reseller). https://floridareading.com/products/noa
47. .lumen glasses. https://www.dotlumen.com/glasses
48. NaviLens. https://www.navilens.com/
49. Microsoft Seeing AI. https://www.microsoft.com/en-us/garage/wall-of-fame/seeing-ai/
50. Google Lookout help. https://support.google.com/accessibility/android/answer/9031274
51. Microsoft Soundscape. https://www.microsoft.com/en-us/research/product/soundscape/
52. May & Walker, bone-conduction audio and environmental localisation (2017). https://pubmed.ncbi.nlm.nih.gov/28237013/
53. Raspberry Pi Zero 2 W product brief. https://pip-assets.raspberrypi.com/categories/584-raspberry-pi-zero-2-w/documents/RP-008359-DS-1-raspberry-pi-zero-2-w-product-brief.pdf
54. Raspberry Pi Pico 2 W datasheet. https://pip-assets.raspberrypi.com/categories/1088-raspberry-pi-pico-2-w/documents/RP-008304-DS-3-pico-2-w-datasheet.pdf
55. Espressif, ESP32-P4. https://www.espressif.com/en/products/socs/esp32-p4
56. Apple, AVCaptureDevice external device type. https://developer.apple.com/documentation/avfoundation/avcapturedevice/devicetype-swift.struct/external
57. Levitt, transformed up-down methods in psychoacoustics (1971). https://doi.org/10.1121/1.1912375
58. Meta, DAT StreamConfiguration reference (v0.9). https://wearables.developer.meta.com/docs/reference/android/dat/0.9/com_meta_wearable_dat_camera_types_streamconfiguration
59. Seeed Studio, XIAO ESP32-C3 Getting Started (400 KB SRAM, 4 MB flash, 1×I2C, 1×SPI, 11 GPIO, no integrated camera, typical 75 mA Wi-Fi-active). https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/
60. AWS Neuron, NxD Inference release notes (Trn2 and later). https://awsdocs-neuron.readthedocs-hosted.com/en/latest/release-notes/components/nxd-inference.html
61. AWS Neuron, Qwen2-VL on Trainium2 tutorial. https://awsdocs-neuron.readthedocs-hosted.com/en/latest/libraries/nxd-inference/tutorials/qwen2-vl-tutorial.html
62. Hugging Face, Optimum Neuron supported architectures. https://huggingface.co/docs/optimum-neuron/en/supported_architectures
63. AWS, EC2 Inf2 instances. https://aws.amazon.com/ec2/instance-types/inf2/
64. AWS, EC2 Trn2 instances. https://aws.amazon.com/ec2/instance-types/trn2/
65. AWS, Lambda arm64 / Graviton2 foundations. https://docs.aws.amazon.com/lambda/latest/dg/foundation-arch.html
66. AWS, Bedrock Nova Lite model card. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-amazon-nova-lite.html
67. AWS, EC2 accelerator instance quotas. https://docs.aws.amazon.com/ec2/latest/instancetypes/ec2-instance-quotas.html